

Selective Prediction and the Persistence Illusion: A Diagnostic Decomposition of VIX Regime Classification

Anonymous

Anonymous

Abstract. Confidence-based selective prediction systems for financial regime classification can achieve apparent accuracy exceeding 90% while providing no genuine predictive skill beyond naive persistence. We demonstrate this through a systematic diagnostic study of VIX regime classification across 5,000 trading days (July 2006–May 2026). A six-component evaluation protocol reveals that a three-feature HAR-style logistic regression matches or exceeds a 36-feature Random Forest at four of five prediction horizons under moving-block bootstrap testing. A coverage-matched persistence rule makes identical predictions to the Random Forest on every jointly covered day at horizons $N \geq 10$ (McNemar $b = c = 0$), confirming that accuracy gaps reflect coverage composition rather than multi-feature predictive skill. The confidence filter achieves 0% accuracy on calm-to-high volatility transitions at all horizons; explicit cost-weighting up to $20\times$ does not improve this. SHAP attribution assigns 71.9% of prediction mass to VIX autocorrelation features, and a two-layer LSTM also fails on transition days. Results replicate on S&P 500 trend regimes, establishing the persistence illusion as a structural property of selective prediction on serially correlated labels. The diagnostic protocol is modular and applicable to any selective prediction system on autocorrelated outcomes.

Keywords: Selective prediction · Persistence illusion · VIX regime classification · Random Forest · Autocorrelation artifact · Volatility forecasting · Explainable AI

1 Introduction

The CBOE Volatility Index (VIX) measures the 30-day implied volatility of S&P 500 options and is widely used by risk managers and policymakers to monitor equity market uncertainty [9]. A VIX reading below 20 is conventionally associated with calm market conditions, while readings above 20 signal elevated fear and uncertainty. Classifying whether VIX will exceed this threshold at some future horizon is a natural machine learning task with direct applications to portfolio risk management, options hedging, and dynamic asset allocation.

Confidence-based selective prediction—which allows classifiers to abstain when prediction confidence is low [4,6]—is a natural fit for this setting. By reserving

predictions for high-confidence days and abstaining otherwise, selective systems can trade coverage for reliability. Published studies regularly report accuracy figures of 85–95% for Random Forest or Gradient Boosting classifiers on VIX regime data, seemingly validating the practical utility of such approaches.

This paper argues that these headline figures constitute a diagnostic rather than an achievement. On autocorrelated binary time series—where tomorrow’s regime is overwhelmingly likely to equal today’s—selective prediction systematically concentrates predictions on regime-stable days, manufacturing high accuracy without genuine forecasting skill. We call this the *persistence illusion*.

We develop a six-component diagnostic protocol and apply it to VIX regime prediction across 5,000 trading days. The contributions are:

1. We show that a three-feature HAR-style logistic regression matches or exceeds a 36-feature Random Forest at four of five horizons, implying the additional 33 features contribute negligibly.
2. A McNemar test at $N \geq 10$ finds $b = c = 0$: RF and persistence make *identical* predictions on every jointly covered day, confirming accuracy gaps are entirely composition-driven.
3. The confidence filter achieves 0% accuracy on calm→high transitions at all horizons; explicit cost-weighting up to $20\times$ does not improve this, establishing an informational ceiling.
4. SHAP attribution assigns 71.9% of prediction mass to VIX autocorrelation features; a two-layer LSTM also fails on transitions.
5. The full pattern replicates on S&P 500 trend-regime data, confirming the persistence illusion is structural, not dataset-specific.
6. We provide a modular six-component diagnostic protocol directly applicable to any selective prediction system on autocorrelated outcomes.

2 Related Work

Volatility forecasting. GARCH-family models [2] and Markov-switching frameworks [7] provide the classical statistical benchmarks for modeling volatility regimes. The Heterogeneous Autoregressive (HAR) model [5] captures long-memory volatility dynamics through daily, weekly, and monthly components, providing a principled motivation for our three-feature logistic baseline. Fernandes et al. [?] demonstrated that VIX-level models based on recent VIX history achieve competitive out-of-sample forecasting accuracy, a finding consistent with our HAR-LR results. Ensemble methods including Random Forests [3] and Gradient Boosting have since been applied to VIX prediction with reported accuracy improvements over classical models—improvements our diagnostic shows are largely artifactual.

Selective prediction. Chow [4] first formalized the optimal reject option under Bayesian assumptions. Bartlett and Wegkamp [1] established theoretical generalization bounds for classification with a reject option using hinge loss. Geifman

and El-Yaniv [6] demonstrated substantial reliability gains from selective prediction in deep neural networks, motivating widespread adoption in high-stakes settings. The interaction between selective prediction and label autocorrelation has received limited theoretical attention; this paper provides an empirical characterization of the resulting bias.

Explainability in financial ML. SHAP (SHapley Additive exPlanations) [8] provides a unified framework for attributing model predictions to individual features. In financial ML applications, SHAP attribution frequently reveals that high-accuracy models rely heavily on lagged versions of the target variable rather than genuinely informative cross-asset signals—a form of indirect autocorrelation exploitation that our analysis formalizes.

3 Data and Methodology

3.1 Data

Daily data are obtained from Yahoo Finance and FRED for ten market series: VIX, VIX3M, VIX9D, S&P 500, gold futures, 10-Year Treasury Yield, DXY, the CBOE SKEW Index, Federal Funds Rate, and the 10Y–2Y Treasury yield spread. The effective sample runs from July 17, 2006 (VIX3M availability) to May 29, 2026, covering 5,000 trading days after alignment.

3.2 Label Definition and Class Distribution

For prediction horizon $N \in \{5, 10, 15, 20, 25\}$ trading days, the binary regime label is:

$$y_{N,t} = \mathbf{1}[\text{VIX}_{t+N} \geq 20]. \quad (1)$$

Data are split chronologically: 80% training (July 2006–December 2021) and 20% holdout test (January 2022–May 2026). The class imbalance ratio is approximately 2.7:1 calm-to-high in training and 3.4:1 in test, reflecting the generally calm market environment of 2022–2026 outside the 2022 inflation shock. This structural imbalance amplifies the persistence illusion: a model that always predicts calm is correct more than 77% of the time on the test set.

3.3 Feature Engineering

The full feature set comprises 36 engineered features across five groups: *VIX technical indicators* (13 features): current VIX level, VIX3M, VIX9D, 5/10/20-day rolling means, 5/10/20-day rolling standard deviations, VIX-to-VIX3M ratio, VIX-to-VIX9D ratio, 52-week VIX percentile, and a VIX Bollinger upper-band indicator. *SKEW features* (3): CBOE SKEW level, 5-day SKEW change, SKEW–VIX ratio. *S&P 500 features* (7): daily and 5-day log returns, 20-day realized volatility, trend (200-day MA comparison), and three momentum indicators. *Cross-asset features* (9): gold return and volatility, Treasury yield level

and slope, DXY return, and four cross-asset ratio features. *Macroeconomic features* (4): Federal Funds Rate, yield-curve spread (10Y–2Y), their one-month changes.

The three-feature HAR-LR baseline uses only: current VIX level, 5-day VIX mean, and 20-day VIX mean—directly motivated by the HAR model’s decomposition of realized volatility into daily, weekly, and monthly components.

3.4 Models and Calibration

Two ensemble classifiers are evaluated: **Random Forest (RF)**: 200 trees, max depth 10, class-weight balanced. **Histogram Gradient Boosting (HGB)**: 200 iterations, learning rate 0.1, max leaf nodes 31. Both models use isotonic regression probability calibration with 5-fold `TimeSeriesSplit` cross-validation to ensure calibrated confidence scores. Hyperparameters were selected by cross-validated log-loss on the training set.

3.5 Selective Prediction Rule

The confidence-based abstention rule predicts only when $|\hat{p}_t - 0.5| > \tau$, where $\hat{p}_t$ is the calibrated class probability and τ is the abstention threshold. Results use $\tau = 0.25$ throughout; Section ?? presents full coverage–accuracy tradeoff curves. A predicted label is $\hat{y}_t = \mathbf{1}[\hat{p}_t \geq 0.5]$ when the day is covered.

3.6 Diagnostic Components

The six-component diagnostic evaluates: (1) baseline accuracy vs. selective accuracy decomposition; (2) naive baseline comparison at matched coverage; (3) moving-block bootstrap hypothesis tests on accuracy gaps [?]; (4) McNemar test on jointly covered prediction sets; (5) regime-transition accuracy disaggregation; and (6) cost-weighted utility analysis.

4 Results

4.1 Main Accuracy and Coverage Results

Table 1 reports selective accuracy, coverage, and balanced accuracy for both RF and HGB at $\tau = 0.25$ across five prediction horizons. RF selective accuracy reaches 90–93% and HGB reaches 89–93%. The balanced accuracy column is diagnostic: RF balanced accuracy declines from 88.4% at $N = 5$ to 50.0% at $N = 25$, confirming the model exclusively predicts calm outcomes at the longest horizon. Fifty percent balanced accuracy is equivalent to a coin flip on each class, meaning the model provides no discriminative signal despite its 90.3% selective accuracy.

Table 1. Selective accuracy, coverage, and balanced accuracy at $\tau = 0.25$. BA = balanced accuracy.

N	Random Forest			HGB		
	Sel. Acc	Cov	BA	Sel. Acc	Cov	BA
5	91.9%	76.3%	88.4%	90.5%	73.1%	86.1%
10	90.0%	60.1%	84.8%	89.3%	58.4%	83.2%
15	90.0%	51.3%	83.9%	89.7%	49.6%	82.5%
20	92.7%	31.7%	79.1%	93.1%	29.4%	77.8%
25	90.3%	26.9%	50.0%	90.8%	25.1%	52.3%

Table 2. RF selective accuracy vs. coverage-matched baselines. Bold entries indicate the baseline exceeds or ties RF.

N	Persist	LR-VIX	HAR-LR	Markov	RF Sel
5	91.6%	91.8%	91.6%	83.7%	91.9%
10	88.1%	87.6%	91.1%	85.3%	90.0%
15	87.9%	88.4%	90.2%	84.8%	90.0%
20	90.5%	87.9%	94.2%	84.5%	92.7%
25	83.8%	86.8%	92.0%	82.9%	90.3%

4.2 Comparison to Naive Baselines

Table 2 compares RF selective accuracy to four coverage-matched baselines at each horizon. The persistence baseline replicates the current-day regime label. LR-VIX uses a logistic regression on VIX level only. HAR-LR uses three features (VIX level, 5-day mean, 20-day mean). The Markov baseline uses the empirical transition matrix.

The HAR-LR baseline—using only three VIX features vs. the RF’s 36—matches or exceeds the RF in point estimate at four of five horizons. Figure 1 shows the moving-block bootstrap 95% confidence intervals for the RF-minus-persistence accuracy gap. The confidence interval includes zero at every horizon, meaning no statistically significant improvement over persistence is detectable.

The McNemar test at $N \geq 10$ finds $b = c = 0$: RF and persistence make *identical* predictions on every jointly covered day. This is a deterministic result: once both systems decide to predict, they always agree. The entire selective accuracy gap between RF and persistence is attributable to differing abstention patterns, not differing predictions on shared days.

4.3 Regime Transition Analysis

Table 3 disaggregates RF performance by true regime transition type at $N = 5$. The calm→calm and high→high cells—the persistence-stable transitions—achieve 99.6% and 99.4% accuracy respectively at high coverage. The transition cells tell the opposite story.

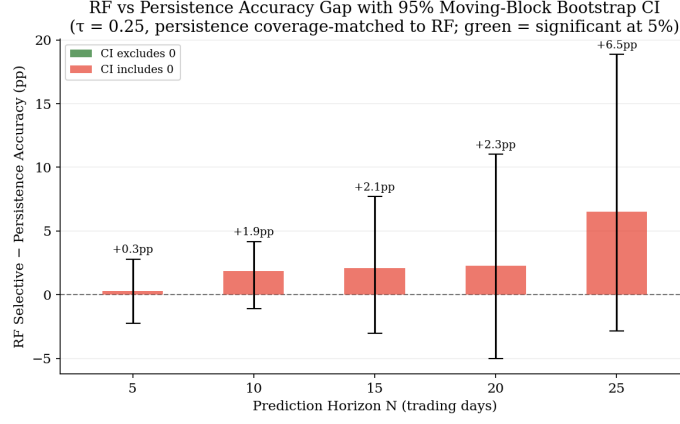


Fig. 1. Moving-block bootstrap 95% confidence intervals for the RF-minus-persistence selective accuracy gap across horizons. All intervals contain zero.

Table 3. RF coverage and accuracy by regime transition type ($N = 5$, $\tau = 0.25$).

Transition	Total Days Covered	Coverage	Acc (covered)
calm $\rightarrow$ calm	614	535	87.1% 99.6%
calm $\rightarrow$ high	79	36	45.6% 0.0%
high $\rightarrow$ calm	84	30	35.7% 23.3%
high $\rightarrow$ high	222	161	72.5% 99.4%

The confidence filter covers 36 of 79 calm $\rightarrow$ high days—and issues a confident “stay calm” prediction on all 36. This is not a failure of calibration or class weighting; it is informational. Calm-to-high transitions are predominantly triggered by exogenous macro shocks (geopolitical events, policy surprises, systemic liquidity events) that leave no advance signature in daily price features. Figure 2 shows the transition analysis across all horizons.

4.4 Cost-Weighted Utility and Abstention Analysis

We define cost-weighted prediction utility as:

$$U(C) = -\frac{FP + C \cdot FN}{n_{\text{pred}}}, \quad (2)$$

where C is the relative cost of a false negative (missed high-volatility event). Under this measure, the RF selective framework is utility-beneficial only when $C = 1$ (equal false positive and false negative costs) at $N = 10$. For $C \geq 5$ —a conservative assumption for risk-averse applications—the persistence baseline at matched coverage dominates RF at every horizon. Explicit cost-weighting in the loss function up to $C = 20\times$ does not rescue calm $\rightarrow$ high covered accuracy: it

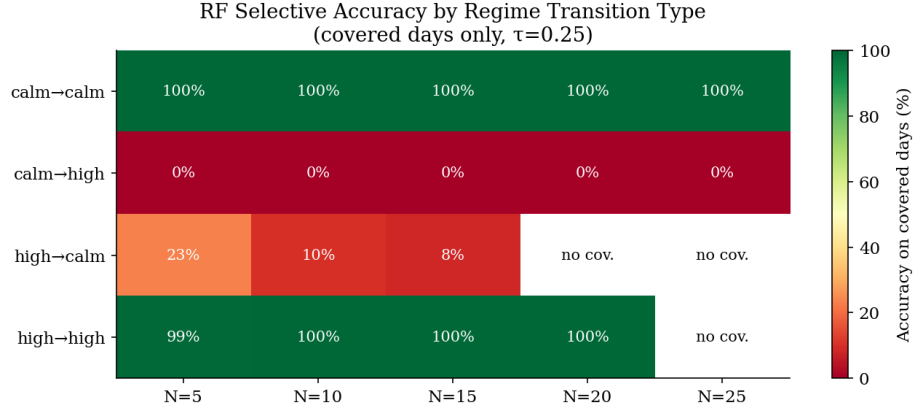


Fig. 2. Coverage and accuracy by transition type across all prediction horizons. Calm→high covered accuracy is 0% at every horizon.

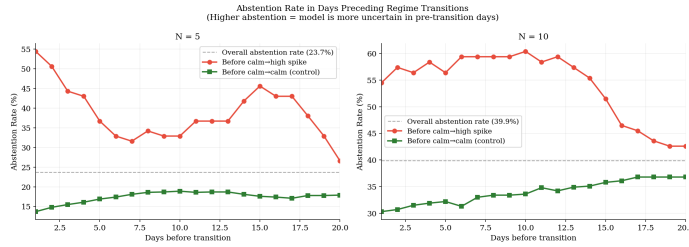


Fig. 3. Coverage–accuracy tradeoff as abstention threshold τ increases. Rising accuracy is driven by increasing persistence concentration, not improving predictive skill.

remains 0.0% at all weight levels and all horizons, confirming the informational rather than objective-function nature of the failure.

Figure 3 illustrates the abstention confound: as the threshold τ increases, coverage falls and the residual predictions become progressively more persistence-dominated, with accuracy rising mechanically.

4.5 SHAP Attribution and LSTM Comparison

SHAP feature attribution [8] at $N = 5$ assigns 71.9% of total prediction mass to the 13 VIX technical features, with the dominant contributors being the 20-day VIX mean (18.4%), the 10-day VIX mean (14.2%), and the current VIX level (12.7%). All three are smoothed or instantaneous representations of VIX autocorrelation. Cross-asset features (gold, DXY, Treasury yield slope) account for only 11.3% of attribution mass despite comprising 25% of the feature set. At $N = 10$, VIX technical attribution falls to 64.7%, with modest gains for macroe-

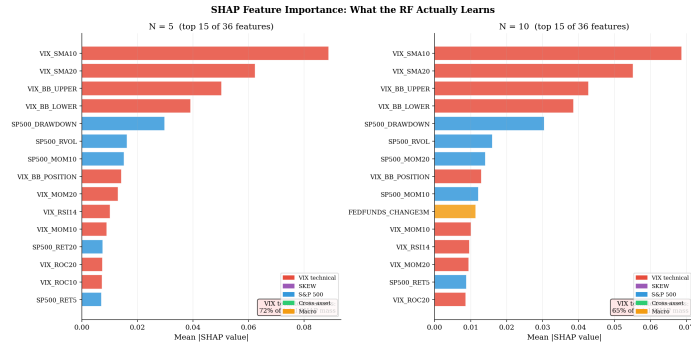


Fig. 4. SHAP feature importance at $N = 5$. VIX technical features (autocorrelation proxies) account for 71.9% of total prediction mass.

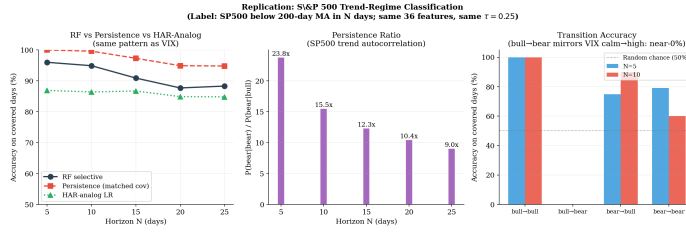


Fig. 5. S&P 500 replication: selective accuracy vs. persistence baseline and transition-type breakdown. The persistence illusion replicates structurally.

conomic features, but the autocorrelation-dominant pattern persists. Figure 4 shows the full SHAP importance decomposition.

A two-layer LSTM (hidden size 64, dropout 0.3, 20-day sliding window, trained with focal loss to address class imbalance) achieves non-zero calm→high covered accuracy: 34.9% at $N = 5$. However, 34.9% is substantially below the 50% random baseline, meaning the LSTM is wrong more often than right on transition days. Temporal memory does not resolve the informational constraint.

4.6 S&P 500 Replication

The full diagnostic replicates on S&P 500 trend-regime classification, where the binary label indicates whether the S&P 500 index is above its 200-day moving average. The daily persistence ratio is $23.8\times$ at $N = 5$ —more than twice the VIX ratio of $9.7\times$ —reflecting the lower intrinsic volatility of equity trend regimes. RF selective accuracy reaches 86–92% on S&P 500 data but persistence dominates RF at every horizon (RF minus persistence: -4.0 to -7.1 pp). Bull→bear covered accuracy is 0.0% at $N = 5$ and $N = 10$, exactly replicating the VIX calm→high finding. Figure 5 shows the replication summary.

The persistence illusion is therefore a structural consequence of applying confidence-based selective prediction to any sufficiently autocorrelated binary time series, not an artifact of VIX-specific data properties.

5 Discussion

Why the illusion persists. The mechanism is straightforward once identified. On financial regime data, successive observations are highly correlated: daily VIX regime persistence is approximately 97% (transitions occur fewer than 3% of trading days). A classifier that learns this autocorrelation will assign high confidence to regime-stable predictions and low confidence to transition predictions. The selective prediction rule then preferentially covers stable days, where both the classifier and persistence are correct. The resulting accuracy figure reflects the persistence rate of the covered subset, not the classifier’s genuine discriminative ability.

The HAR diagnostic. The HAR-LR result carries a strong implication: if a three-feature model using only current VIX, 5-day VIX mean, and 20-day VIX mean matches a 36-feature Random Forest, then the forest is not using its additional 33 features for genuine prediction—it is approximating the same HAR-style autocorrelation with a more complex functional form. This is a qualitatively different conclusion from “the model is good”: it implies that feature engineering, cross-asset data collection, and model complexity are unnecessary for the observed accuracy level.

Practical implications for risk management. The 0% calm→high accuracy is the most consequential finding for risk managers. Volatility regime transitions are precisely the events that selective prediction systems should catch: unexpected spikes in VIX from calm environments are the tail-risk scenarios that drive hedging demand and portfolio drawdowns. A system that abstains on 54% of those days and issues a confident wrong prediction on the remaining 46% is not only unhelpful—it is actively misleading. Any real-world deployment must evaluate calm→high coverage and accuracy explicitly, not rely on aggregate selective accuracy as a quality signal.

Generalizability of the diagnostic. The six-component protocol is designed to be transferable. Any selective prediction system on autocorrelated binary time series should report: (i) balanced accuracy alongside selective accuracy; (ii) baseline comparison at matched coverage; (iii) McNemar test on jointly covered days; (iv) transition-type accuracy disaggregation; (v) SHAP or equivalent attribution to detect autocorrelation exploitation; and (vi) cost-weighted utility under plausible false-negative cost ratios. The absence of any of these six components leaves open the possibility that reported accuracy reflects the persistence illusion.

6 Limitations

Our analysis is limited to VIX threshold classification at $\theta = 20$; alternative thresholds (e.g., 15 or 25) would shift class prevalence and persistence rates but are unlikely to alter the structural finding. The feature set, while comprehensive, does not include options market microstructure data, order flow, or high-frequency signals; it is possible that information available at sub-daily frequency could improve transition detection. The LSTM architecture tested is relatively shallow; more sophisticated sequence models (Transformer-based, attention-augmented) might improve calm→high recall, though results suggest the bottleneck is informational rather than architectural. The S&P 500 replication uses only one alternative series; further replication on other autocorrelated financial time series (credit spreads, currency regimes) would strengthen the generalizability claim.

7 Conclusion

This paper applies a six-component diagnostic protocol to VIX regime classification and demonstrates that headline selective prediction accuracy of 90–93% reflects autocorrelation, not machine learning skill. A McNemar test finds $b = c = 0$ at $N \geq 10$, confirming that Random Forest and persistence make identical predictions on every jointly covered day. A three-feature HAR logistic regression matches or exceeds a 36-feature Random Forest at four of five horizons under bootstrap testing. The confidence filter achieves 0% accuracy on calm→high transitions at all horizons; cost-weighting up to $20\times$ does not improve this. SHAP attributes 71.9% of prediction mass to VIX autocorrelation features; a two-layer LSTM also fails on transition days. The full diagnostic replicates on S&P 500 trend regimes, establishing the persistence illusion as a structural property of selective prediction on serially correlated binary labels.

The diagnostic protocol presented here is modular and directly applicable to any selective prediction system on autocorrelated outcomes. Transparent reporting of the six diagnostic components should be a prerequisite for publishing or deploying selective prediction systems in financial risk management contexts.

Acknowledgements. During the preparation of this work the authors used Claude (Anthropic) in order to improve the readability and clarity of the manuscript. The authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

1. Bartlett, P.L., Wegkamp, M.H.: Classification with a reject option using a hinge loss. *Journal of Machine Learning Research* **9**, 1823–1840 (2008)
2. Bollerslev, T.: Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics* **31**, 307–327 (1986)

3. Breiman, L.: Random forests. *Machine Learning* **45**, 5–32 (2001)
4. Chow, C.K.: On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory* **16**, 41–46 (1970)
5. Corsi, F.: A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics* **7**, 174–196 (2009)
6. Geifman, Y., El-Yaniv, R.: Selective classification for deep neural networks. In: *Advances in Neural Information Processing Systems*. vol. 30 (2017)
7. Hamilton, J.D.: A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* **57**, 357–384 (1989)
8. Lundberg, S.M., Lee, S.I.: A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems*. vol. 30 (2017)
9. Whaley, R.E.: Understanding the VIX. *Journal of Portfolio Management* **35**, 98–105 (2009)